How Our Landing-Site Algorithm Works

Step-by-step design for the team
NASA Space Apps Challenge 2026: Lunar Landing-Site Challenge

Ash
Princess Sumaya University for Technology

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1 What the Algorithm Is Based On

We do not invent the science. The algorithm combines three things that already exist:

1. **Physics.** Sunlight and radio travel in straight lines, and the Moon has no atmosphere to bend them. So "can this spot see the Sun?" is exactly "does the straight line from here to the Sun hit any terrain?"
2. **NASA's measured data.** Terrain heights come from the LOLA laser altimeter on the Lunar Reconnaissance Orbiter. Sun and Earth positions for any hour come from JPL's ephemerides through the SPICE toolkit [4].
3. **NASA's requirements.** They define what "good" means: land within 6° of the pole, slope under 10° over 100 m, direct line of sight to Earth with no relay, and enough sunlight [1].

NASA scientists use this same "horizon method" to publish long-term illumination and Earth-visibility maps of the poles [2]. That proves the method is sound and gives us data to check against (Step 5). Figure 1 shows the full pipeline.

2 Step 0: Prepare the Terrain and Candidate Sites

- **Get the terrain.** Download two LOLA south-pole elevation maps [3]: one with 20 m pixels for the terrain near each site, and one with 240 m pixels for distant terrain. Step 1 explains why both are needed.
- **Crop with a margin.** Candidate sites lie within 6° of the pole, but Step 1 looks far beyond each site. The 240 m map must therefore reach the maximum look-out distance $D_{\max} = 250$ km beyond every site, and the 20 m map must reach 40 km beyond every site.
- **Fix the coordinates.** Both maps use the same polar-stereographic projection. A site is given by its map position x, y in metres, plus its lat/lon. A pixel (row, column) is not a unique address, because the two maps have different pixel sizes.
- **Slope gate.** Compute the slope over a 100 m baseline and drop every site at 10° or steeper.
- **Candidate grid.** Start with every k-th pixel, then refine around the best areas. NASA's published average illumination map [2] can pre-select promising areas so we do not waste time on permanently dark craters.

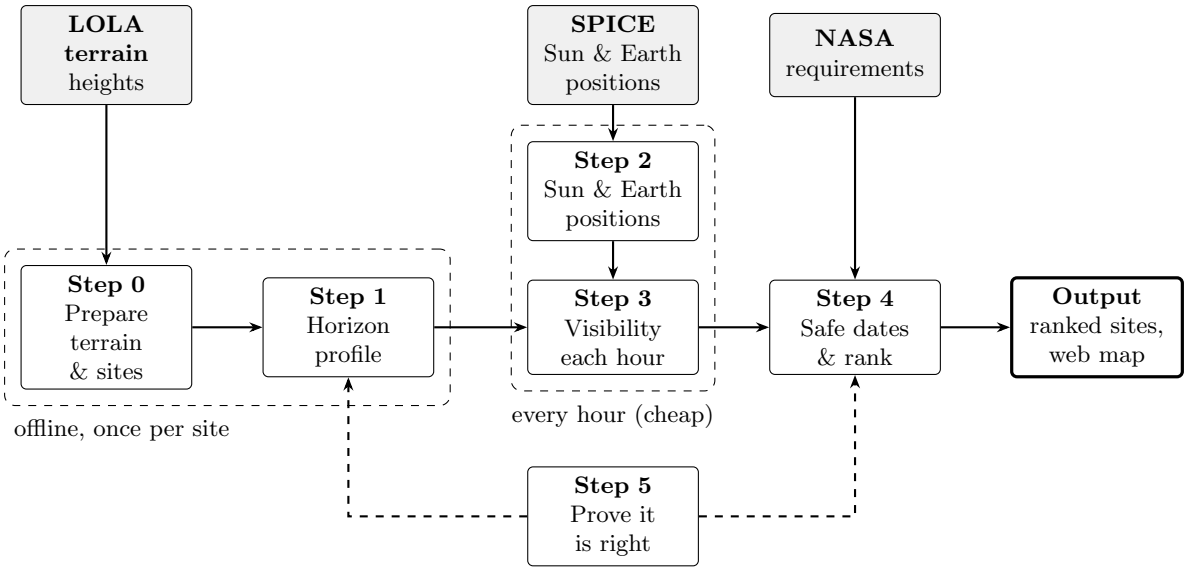


Figure 1: The pipeline. Shaded boxes are inputs from NASA. Steps 0–1 run once, offline; Steps 2–3 are a fast lookup for any dates the user picks. Step 5 checks the results of Steps 1 and 4.

Direction convention (critical). At the south pole every direction is "north", so compass bearings do not work. All steps measure azimuth θ from the map's $+x$ axis, counterclockwise, 0–360°. Longitude 0 is map up ($\theta = 90°$), so Earth always sits near $\theta = 90°$. Steps 1 and 2 must use exactly this convention.

3 Step 1: Horizon Profile (the Hardest Step)

Stand at a site and look in direction θ. Walk outward over the terrain. At each distance d, compute the angle up to the terrain. The steepest angle found is the horizon $H(\theta)$ in that direction (Figure 2). Repeat for 1,440 directions, 0.25° apart, to get the site's full skyline.

$$H(\theta) = \max_{d \le D_{\max}} \operatorname{atan2}\Big((R+z)\cos\varphi - (R+h_0+m),\ (R+z)\sin\varphi\Big), \qquad \varphi = d/R \tag{1}$$

Here $z(d)$ is the terrain height at distance d (interpolated between pixels), h_0 is the site height, m is the mast height, $R = 1737.4\,\text{km}$ is the Moon's radius, and φ is the angle between the site and the terrain point, seen from the Moon's centre. The formula is exact on a sphere. For intuition, it is almost the same as $\arctan\big((z - h_0 - m - d^2/2R)/d\big)$, where $d^2/2R$ is how far the Moon's curved surface drops below a flat line. The curvature matters: at 20 km the drop is about 115 m, which shifts the angle by about 0.3°. That is significant when the Sun is only about 1.5° up.

Two terrain maps. Within 40 km of the site the walk samples the 20 m map; beyond that it samples the 240 m map. The near field decides more than it seems. On the 240 m map alone, our test site on Shackleton's rim looks sunlit about 87% of the time. On 20 m terrain, the same site seen from a 2 m mast is almost always dark (0.3%), because a crest 103 m away and 9.3 m higher blocks the Sun. A 10 m mast lifts it back to about 80%. The 240 m map had smoothed that crest away.

True great circles. Each walk follows the great circle that leaves the site in direction θ, at true surface distance, and every sample is projected onto the map. An earlier version walked straight lines on the map instead. That was fine near the pole (0.003° off at our rim site) but

up to 0.05° off for sites 5–6° out, more than our 0.02° tolerance. The engine now agrees with an independent spherical-trigonometry walk to within 10^{-6} degrees at every distance from the pole.

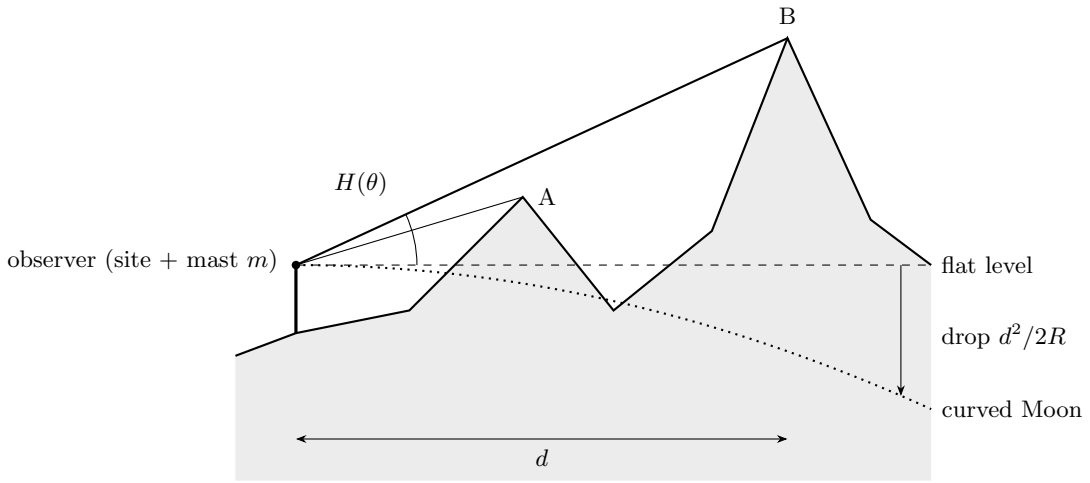


Figure 2: Step 1 in one direction. Keep the steepest angle to the terrain. Peak A is nearer; peak B is farther but higher, and it sets the horizon because its angle is larger. Heights and curvature are exaggerated.

Algorithm 1 Horizon profile for one site

Require: terrain Z, site (x_0, y_0), height h_0, mast m, look-out distance $D_{\max}$

for $\theta = 0°, 0.25°, \ldots, 359.75°$ **do**
  $H[\theta] \leftarrow -90°$; $d \leftarrow d_{\min}$
  while $d \le D_{\max}$ **do**
    $(x, y) \leftarrow (x_0 + d\cos\theta,\ y_0 + d\sin\theta)$
    $z \leftarrow \textsc{Interpolate}(Z, x, y)$ ▷ 20 m map within 40 km, 240 m beyond
    $\alpha \leftarrow$ angle to (d, z) from Equation (1)
    $H[\theta] \leftarrow \max(H[\theta], \alpha)$
    $d \leftarrow d + \textsc{Step}(d)$ ▷ $\max(\text{pixel}/2,\ 0.002\,d)$
  end while
end for
return H

Why compute the horizon first. Checking the Sun and Earth directly would need two walks over the terrain for every hour: 17,520 walks per site for a year. The horizon needs 1,440 walks once, and after that every hour of every year is a lookup. That is about 12 times less work for one year, more for every extra year, and trying new dates costs almost nothing.

What makes it hard. The total cost is (sites) × (mast heights) × 1,440 × (steps per walk). The step grows with distance, which gives 2,458 steps out to 250 km. With 10,000 sites and 4 mast heights (0, 2, 5 and 10 m, so Step 4 can recommend a mast), that is about 1.4×10^{11} terrain samples. Our engine handles this in Numba. It runs sites in parallel, compares tangents instead of angles, and stops each walk once even the highest terrain on the map could no longer rise above the horizon found so far. It matches a simple reference version to within 10^{-6} degrees. It does 400 sites × 4 masts on 20 m + 240 m terrain in 10 seconds, and the full grid is estimated at about 5 minutes.

4 Step 2: Sun and Earth Positions

For every site and every hour, use SPICE (through the SpiceyPy library [6]) to get the direction of the Sun and Earth as seen from that site: azimuth $\mathrm{az}(t)$ in our map convention, and elevation $\mathrm{el}(t)$ above the local flat level. The kernels needed are the leap-second file, a planetary ephemeris (for example DE440), and the Moon's orientation kernels.

The Sun and Earth move very differently (Figure 3):

- **Sun:** goes all the way around the horizon once a month, staying within about $\pm 1.5°$ of level. So sunlight depends on the skyline in *every* direction.
- **Earth:** stays in one narrow direction and bobs up and down by about $\pm 6.7°$ over about 27 days. So Earth contact depends on the skyline in *one* direction only.

These ranges double as sanity checks. If SPICE output falls outside them, something is wrong.

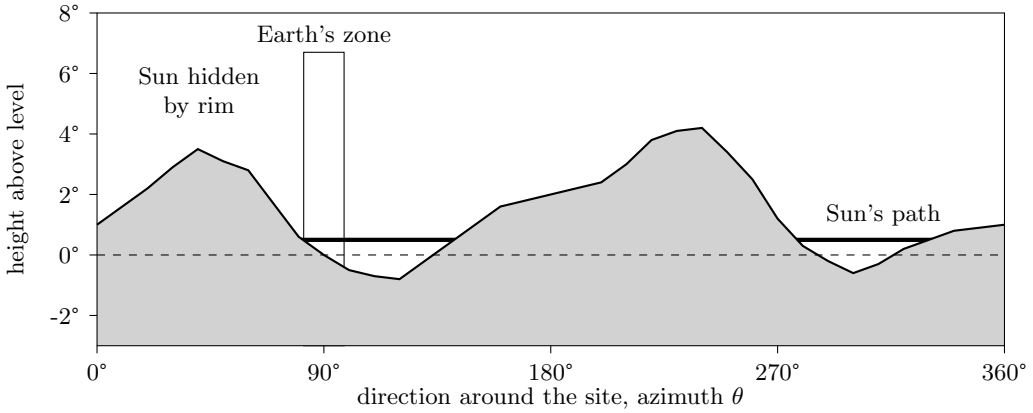


Figure 3: Steps 2 and 3 in one picture (illustrative terrain). Gray is the horizon profile from Step 1. The Sun moves along the flat line and is visible only where the line is above the gray. Earth stays inside the narrow band around 90° (toward longitude 0) and moves up and down; it is visible when it rises above the gray there.

5 Step 3: Visibility Each Hour

For each site and hour t, compare the sky positions with the horizon. H is interpolated between neighbouring directions (0.25° apart). The Sun and Earth are discs, not points (radius $r_\odot \approx 0.27°$, $r_\oplus \approx 0.95°$ as seen from the Moon):

$$s(t) = \mathrm{clamp}\left(\frac{\mathrm{el}_\odot(t) - H(\mathrm{az}_\odot(t)) + r_\odot}{2\,r_\odot},\ 0,\ 1\right) \qquad \text{fraction of the Sun visible (power)} \qquad (2)$$

$$e(t) = \left[\,\mathrm{el}_\oplus(t) + r_\oplus > H(\mathrm{az}_\oplus(t))\,\right] \qquad \text{any part of Earth visible (comms)} \qquad (3)$$

$$b(t) = \left[\,s(t) > 0.5\,\right] \wedge e(t) \qquad \text{power and comms together} \qquad (4)$$

The "any part of Earth" rule matches the one NASA uses for its Earth-visibility maps [2]. Figure 4 shows what Step 3 produces for one site.

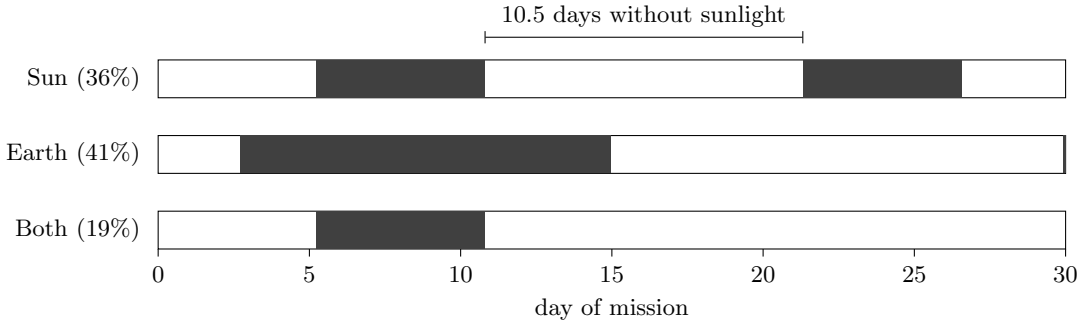


Figure 4: Step 3 output for one site over 30 days (illustrative). Black means visible. Sun and Earth are each visible about 40% of the time, but both together only 19%, and there is a 10.5-day stretch without sunlight that the battery must survive. A monthly average hides all of this.

6 Step 4: Safe Landing Dates and Ranking

Gates. A site must be within 6° of the pole and have a slope under 10° *over 100 m*, measured on the 20 m map. Sites that fail are removed, not ranked. (Until 13 November the demo grid's slope came from two pixels of the 240 m map, a 480 m baseline, which let five too-steep sites through; it was found in a risk review and fixed.)

No weighted score. An earlier version added the metrics into one score with weights. It gave many ties and called dates "best" on which the lander died, so it was removed. Every number below is a physical quantity.

Metrics over a mission of T hours (14 days), for a mission starting at *every hour* from 2026 to 2030: f_{sun}, f_{earth} and f_{both} are the fractions of hours with $s > 0.5$, e and b true; g is the longest stretch without sunlight (the cold soak).

Battery survival. A simple energy budget with panel power P_{panel}, load P_{load} and time step Δt:

$$E_{t+1} = \min\bigl(E_{\max},\ E_t + (P_{\text{panel}}\, s(t) - P_{\text{load}})\,\Delta t\bigr) \tag{5}$$

The lander survives if $E_t \geq E_{\min}$ at every hour. *Battery needed* is $E_{\max}$ minus the lowest level, taken from this simulation, so sunlight that returns partway counts.

Safe landing hour. A landing hour is safe if all three hold:

1. *touchdown*: more than half the Sun and Earth are in view in the landing hour;
2. Earth is in view for at least half of the mission (direct-to-Earth, no relay [1]);
3. the battery never drops below $E_{\min}$.

A landing day is safe if any of its hours is. The headline number per site and mast is the count of *safe landing days*.

Landing accuracy. A lander cannot hit an exact pixel; NASA's reference is within 100 m of the target [1]. So a date counts for a site only if it is safe at every one of 17 points within 100 m of it (the site, 8 points at 50 m and 8 at 100 m). Each of the 17 points must also have ground under 10° over 100 m, since the lander may come down on any of them. With this rule, no site of the demo grid at a 2 m mast is safe within 100 m; at 10 m, the Shackleton rim site has 337 safe landing days and 315 of them hold within 100 m.

Ranking. One list per mast height: the most safe landing days within 100 m first, then the most on the site's own point, then the least battery needed. Sites with none within 100 m stay in the list, below the others.

Mast height. Using the horizons from Step 1 at several mast heights, report the shortest mast with at least one safe landing date.

When to land. Each site gets its best landing hours: a *firm* touchdown first (the Sun's centre and the top of Earth's disc both at least 0.5° above the skyline, so small terrain errors cannot hide them), then the most hours of power and Earth together, then the shortest darkness, then the earliest.

Failsafe. For the chosen landing, the pipeline also reports what goes wrong if it does not go to plan: how many hours late it can land and stay safe (and a one-lunar-month delay), which of the 17 points fail and why, when Earth is in view to upload and confirm the plan, whether the battery lasts through each dark spell, the same mission with half the battery, and how many minutes sunset and Earthset move if the skyline is 0.02° off.

7 Step 5: Prove It Is Right

1. **Smooth-Moon test.** On a perfectly smooth sphere, an observer at height h must see the same horizon in every direction: $H = -\arccos\big(R/(R+h)\big)$, about $-0.087°$ for $h = 2\,\text{m}$. This also checks that $D_{\max}$ reaches past the tangent distance $\sqrt{2Rh} \approx 2.6\,\text{km}$.
2. **Single-cone test.** One cone-shaped peak of height P at distance L must give exactly the angle of Equation (1) for $z = P$, $d = L$ in its direction, blocked at exactly L.
3. **Settings by convergence.** Make every setting finer until the answer stops changing. The rule: every site's skyline stays within 0.02° of the finest settings in 99% of its directions. Each finest setting lies beyond the value we use: 300 km, 11,520 directions, step factor 0.00025, and 60 km of 20 m terrain. On the 240 m map we tested the rim site plus 500 random sites within 5° of the pole (300 km rays from farther out leave the map); on 20 m terrain, the rim site plus 150 random sites within 13 km of the pole. Results:
   - Look-out distance $D_{\max} = 250\,\text{km}$. At 200 km the worst site is still 0.68° off in 99% of its directions; 250 km and 300 km give exactly the same skylines.
   - 1,440 directions. With 360 directions, the worst site is 0.066° off.
   - Step factor 0.002: far from the site, each step is 0.2% of the distance.
   - 20 m terrain out to 40 km. At 20 km, sectors are 0.044° off.
4. **Compare with NASA's maps.** NASA's illumination maps [2] are averages over 18.6 years of hourly steps, so we compute ours the same way: our own sky (SPICE, every hour for 18.6 years) and the Step 3 formulas. We ran the engine at the centres of the 100 NASA 60 m pixels around each of two sites, with 20 m terrain nearby and 240 m beyond. From a 2 m mast (the best fit at both sites; NASA states no observer height):
   - Shackleton rim: Sun correlation 0.92, average difference +0.1 points, 94% of pixels within 10 points; Earth 0.85, +0.6, 89%.
   - Shackleton–de Gerlache ridge: Sun 0.83, +1.9; Earth 0.90, +2.2.

   NASA's maps use the same LOLA terrain, so this checks the method. Always compare at pixel centres, never "NASA's value at the site": a site can sit exactly on the corner of four NASA pixels (at our rim site they read 0%, 6%, 47% and 56%).
5. **Check the sky against NASA JPL Horizons.** Horizons [5] (listed in the challenge's resources) gives the Sun and Earth seen from any point on the Moon. At six named sites, ten hours over 2026–2030 plus the IM-1 and IM-2 landing hours, all 124 positions agree

with our Step 2 to within 0.0003° (the Sun's radius is 0.27°). An independent check with Skyfield agrees to 0.00032°.

6. **Check the terrain against NASA Moon Trek's 5 m map.** At the demo site (the Shackleton rim site, 10 m mast) the mission gets the same verdict on Moon Trek's 5 m LOLA terrain as on our 20 m map. At another site, 2 km from the pole, the two maps differ (a median of 4 m, and a 7–8 m rise 20 m from the site that only the 5 m map shows), and the verdict flips: terrain finer than the map can change the answer, which is why the demo uses a site that passes on both.

8 Files Passed Between Steps

Everyone builds against fake files in these formats from day one, so nobody waits for anyone. Angles are in degrees, times are hourly UTC, and azimuth follows the convention in Step 0.

File	Contents	Shape	Made by	Used by
`dem.npy`	Terrain heights (m) on the map grid	rows × cols	Step 0	Step 1
`sites.csv`	id, x, y, lat, lon, height, slope	one row per site	Step 0	Steps 1, 4
`horizon.npy`	Horizon profile per site and mast height	sites × masts × 1440	Step 1	Step 3
`blocker.npy`	Distance to the terrain that sets the horizon (m)	sites × masts × 1440	Step 1	Step 3, web app
`complete.npy`	True where the walk reached $D_{\max}$ on the map	sites × masts × 1440	Step 1	Step 3
`horizon_meta.json`	Settings: directions, convention, masts, $D_{\max}$, step factor, R, terrain files	—	Step 1	Step 3
`sky.npz`	Sun and Earth az/el each hour	sites × hours	Step 2	Step 3
`visibility.npz`	$s(t)$, $e(t)$, $b(t)$	sites × hours	Step 3	Step 4
`results.json`	Safe landing days, metrics, best dates, one ranking per mast	per site	Step 4	Web app

Table 1: Data contracts between steps. Read the number of directions and the mast heights from `horizon_meta.json` instead of hard-coding them.

9 Who Owns Which Step

Step	Role	Member
Step 0: terrain and sites	Terrain and sites	_____
Step 1: horizon profile	Horizon engine	_____
Step 2: Sun and Earth positions	Sky (SPICE)	_____
Steps 3–4: visibility and scoring	Scoring and mission model (integrator)	_____
Display	Web app	_____
Step 5 and submission	Validation, story and submission	_____

Table 2: Step ownership. The horizon engine owner also writes the smooth-Moon and cone tests.

10 What Makes Our Tool Different

NASA already publishes average illumination and Earth-visibility maps [2], and judges may know them. Averages cannot answer a mission's real questions, and Figure 4 shows why. Our tool answers them for a specific mission:

- When will this site have power and Earth contact *at the same time*?
- Will a lander arriving on a given date survive its whole mission?
- How tall must the mast be at this site?
- Is the site still good if we land 100 m off target?
- Which arrival date is best?

References

[1] NASA, *Artemis Overview, 2025 International Space Apps Challenge*, NTRS 20250009323. https://ntrs.nasa.gov/api/citations/20250009323/downloads/ArtemisOverview2025IntSPAPPSChallenge.pdf

[2] NASA Goddard Planetary Geodesy Data Archive, *Lunar Polar Illumination*. https://pgda.gsfc.nasa.gov/products/69

[3] NASA PDS Geosciences Node, *LRO LOLA polar elevation maps* (LDEM_875S_20M, LDEM_75S_240M). https://pds-geosciences.wustl.edu/lro/lro-l-lola-3-rdr-v1/lrolol_1xxx/data/lola_gdr/polar/

[4] NASA NAIF, *The SPICE Toolkit*. https://naif.jpl.nasa.gov/naif/

[5] NASA JPL Solar System Dynamics, *Horizons System*. https://ssd.jpl.nasa.gov/horizons/

[6] SpiceyPy, Python wrapper for SPICE. https://github.com/AndrewAnnex/SpiceyPy